

L1.7a Implicit Differentiation

§2.6 — what you need to know

The procedure

① **Differentiate both sides.** ② **Divide by dx .** ③ **Collect y' and solve.**

In differentials — what is actually going on:

$$d(x^2 + y^2) = d(25) \implies 2x dx + 2y dy = 0 \implies 2x + 2y y' = 0 \implies \boxed{y' = -\frac{x}{y}}$$

Once you have seen why, you can skip the middle step:

$$\frac{d}{dx}(x^2 + y^2) = \frac{d}{dx}(25) \implies 2x + 2y y' = 0 \implies \boxed{y' = -\frac{x}{y}}$$

Once y' is a single fraction:

Horizontal tangent $\rightarrow m = 0 \rightarrow \text{top} = 0$ **Vertical tangent** $\rightarrow m = \infty \rightarrow \text{bottom} = 0$

Either one gives you **one coordinate, not a point**. Put it back into the *curve's* equation to get the other. Same move for any slope: want the tangent parallel to $y = x$? Set $y' = 1$ and back-substitute.

IMPLICIT CURVES AND LEVEL SETS

$x^2 + y^2 = 25$ is not a function. Read the left side as a function of **two** variables, $S(x, y) = x^2 + y^2$ — the curve is the set of points where S takes the value 25. **A relation is a level set.**

A step (dx, dy) changes S by $dS = 2x dx + 2y dy$, which throws you *off* the curve. At $(3, 4)$ with $dx = dy = -0.1$: $dS = -1.4$, so $S \rightarrow 23.6$ — **not on the circle**.

$$\boxed{d(S) = 0 \text{ on the tangent line}}$$

The tangent is the one direction out of the point that keeps S at the same level. Setting $d(S) = 0$ and solving for dy/dx is implicit differentiation.

CHAIN RULE ON EVERY y -TERM

$$\frac{d}{dx}(y^2) = 2y \frac{dy}{dx} = 2y y' \qquad \frac{d}{dx}(x^2) = 2x \frac{dx}{dx} = 2x$$

Both are the same chain rule. Dividing by dx leaves $\frac{dy}{dx} = y'$ on one and $\frac{dx}{dx} = 1$ on the other.

$$y^3 \rightarrow 3y^2 y'$$

$$xy \rightarrow xy' + y$$

$$\sin y \rightarrow (\cos y) y'$$

power rule, then chain — the inner function is y

a **product** of two things that both vary

same chain rule, trig outer function

TANGENT LINES

Evaluate y' at the point before doing anything else. Once it is a number, the rest is line algebra.

$$x^2 + y^2 = 25 \text{ at } (3, 4)$$

$$y' = -\frac{x}{y} = -\frac{3}{4}$$

$$y = -\frac{3}{4}x + \frac{25}{4}$$

$$x^3 + y^3 = \frac{9}{2}xy \text{ at } (2, 1)$$

$$y' = \frac{\frac{9}{2}y - 3x^2}{3y^2 - \frac{9}{2}x} = \frac{5}{4}$$

$$y = \frac{5}{4}x - \frac{3}{2}$$

y' contains y , and that is a **finished** answer. A relation has more than one y for a given x , so the slope has to depend on which one you are standing on: at $x = 3$ the circle has $(3, 4)$ with slope $-\frac{3}{4}$ and $(3, -4)$ with slope $+\frac{3}{4}$.

HORIZONTAL AND VERTICAL TANGENTS OF AN ELLIPSE

On $2x^2 + y^2 - 4x + 2y + 2 = 0$: $4x + 2yy' - 4 + 2y' = 0$, so $y'(2y + 2) = 4 - 4x$ and $y' = \frac{2 - 2x}{y + 1}$.

Horizontal — top = 0

$$2 - 2x = 0 \Rightarrow x = 1$$

$$\text{back-substitute: } y^2 + 2y = 0$$

$$y(y + 2) = 0$$

$$(1, 0) \text{ and } (1, -2)$$

Vertical — bottom = 0

$$y + 1 = 0 \Rightarrow y = -1$$

$$\text{back-substitute: } x^2 - 2x + \frac{1}{2} = 0$$

$$\text{Loh: } m = 1, d = \sqrt{1 - \frac{1}{2}} = \frac{1}{\sqrt{2}}$$

$$\left(1 + \frac{1}{\sqrt{2}}, -1\right) \text{ and } \left(1 - \frac{1}{\sqrt{2}}, -1\right)$$

Back-substitution is the step that gets skipped. $x = 1$ is a *line*; it meets the curve twice. Only the curve's own equation knows where.

WATCH OUT

✗ $\frac{d}{dx}(y^2) = 2y \rightarrow$ the inner derivative is missing — it is $2y y'$

✗ $\frac{d}{dx}(xy) = y' \rightarrow xy$ is a product — it is $xy' + y$

✗ an answer with y in it is unfinished $\rightarrow$ a relation has several y per x , so the slope must depend on the branch

✗ a vertical tangent means $y' = 0 \rightarrow$ vertical is $m = \infty$ — the bottom is zero

✗ solving top = 0 gives the point $\rightarrow$ it gives one coordinate; the curve's equation supplies the other

✗ implicit differentiation is a new rule $\rightarrow$ it is d of both sides — the same differentials as last lesson